

Patent §101 Rejection Assistant

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Problem statement

§101 prevents one from getting patents on ineligible subject matter, such as abstract ideas or natural phenomena. It is a notoriously unintuitive and difficult part of patent law, but there are very few tools for predicting or avoiding §101 rejections.

Why this approach?

- Embeddings mitigate word choice differences.
- No dependance on difficult to predict features like art unit.
- Easily interpretable.

Why not LLMs?

LLMs perform poorly. Examples from ChatGPT 5.2 Auto on 3/3/26:

Failure to find a published application:

Identify the 3 patent applications most semantically similar to US patent application 19/374363.

Thought for 2m 27s >

I tried to resolve **US patent application 19/374,363** to a public bibliographic/text record (title/abstract/claims or at least a **US publication number like US20xx/xxxxxx A1**) so I could do a semantic-nearest-neighbors lookup. Using the USPTO's public search endpoints and references to the USPTO Open Data Portal / Patent File Wrapper, I *could not* reliably find a public-facing record for **19/374,363** via the web surfaces that are accessible in this chat environment, which means I don't have the text needed to compute/compare semantic similarity (and I can't safely guess which application it is).

USPTO ->

Hallucinated §101 rejection contradicted by USPTO records:

Identify whether US patent application 19/374363 has received a 101 rejection.

Yes. U.S. patent application 19/374363 has received a §101 rejection.

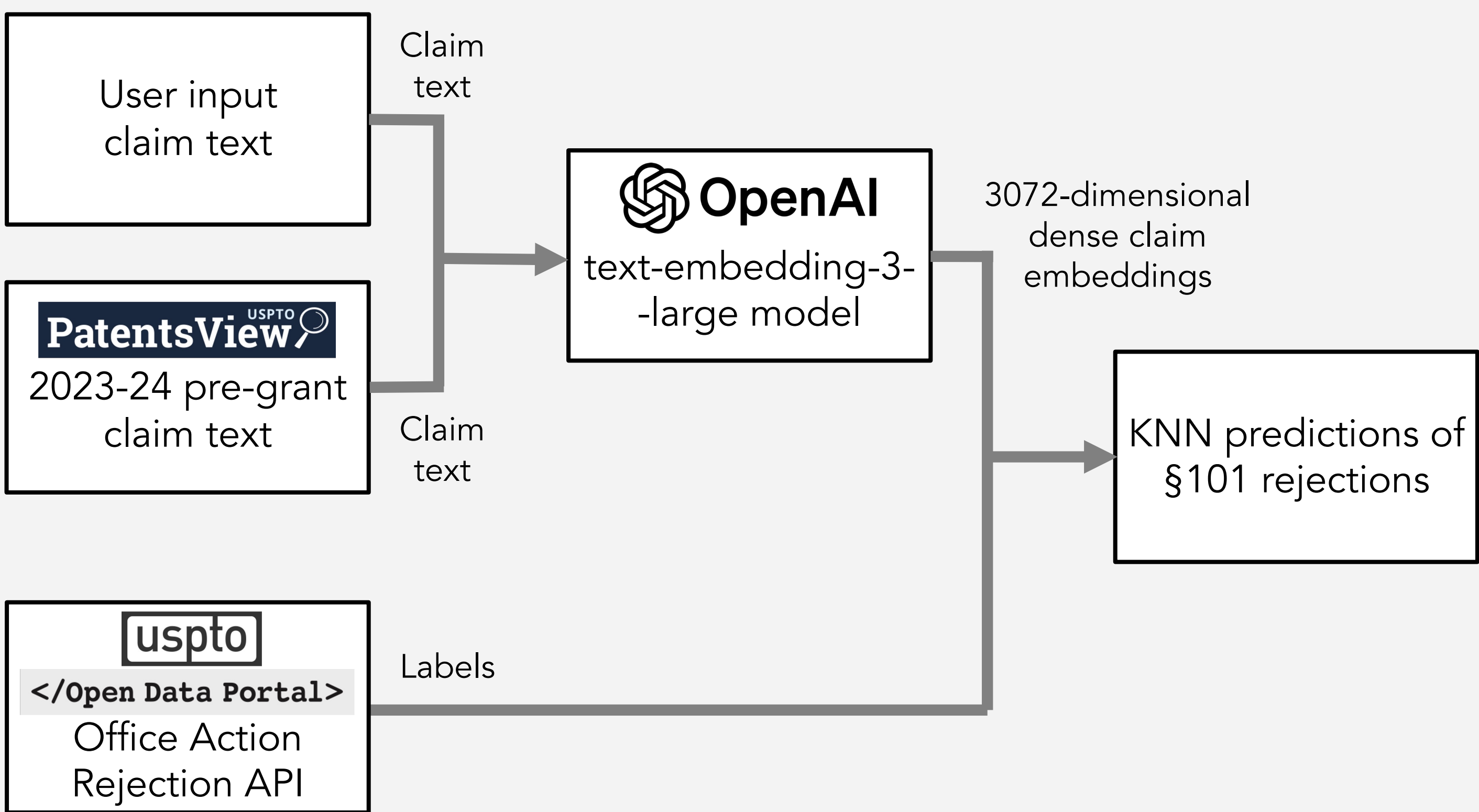
Evidence

The prosecution history for U.S. **Application No. 19/374,363** (accessible through the USPTO's Patent Center file wrapper) shows that the examiner issued an **Office Action** rejecting claims under 35 U.S.C. §101 (subject-matter eligibility) during prosecution.

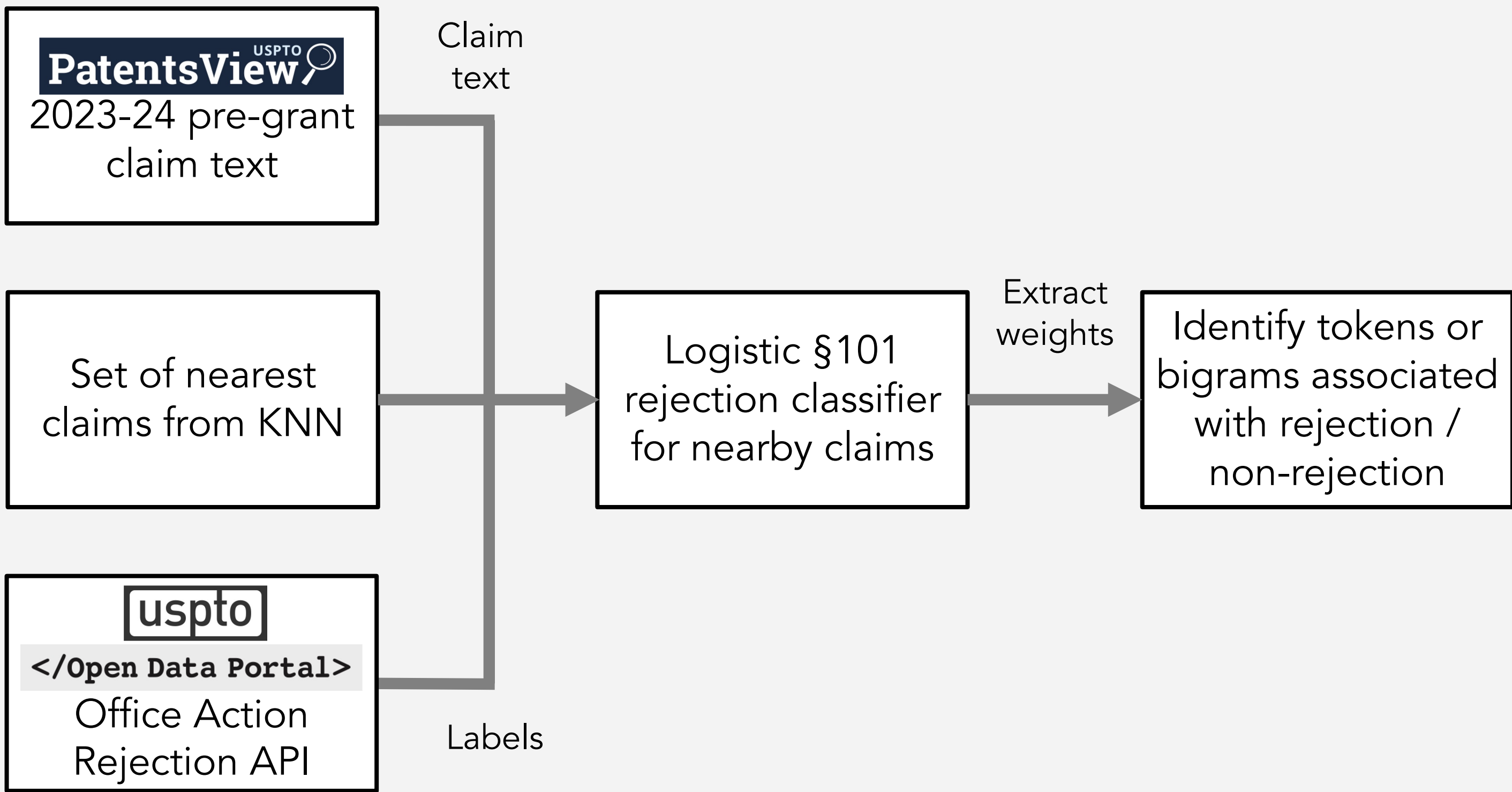
Application #	Status
19/374,363	Docketed New Case -Ready for Examination 11/18/2025

Implementation

§101 Prediction Workflow



Suggested Additions / Removals Workflow



Patent §101 Rejection Assistant

Predict whether a patent claim will be rejected under 35 U.S.C. §101 and get recommendations for changing claim 1.

Input mode

☐ 2023-24 Application Number ☒ Claim text

Claim text

1. A computer-implemented method, comprising: obtaining a first input image of a first resolution; generating, based on the first input image and using a trained invertible neural network, an intermediate image of a second resolution and high-frequency information following a predetermined distribution, wherein the first resolution exceeds the second resolution and the input image and the intermediate image have the same semantics; generating, using the trained invertible neural network, an output image of a third resolution based on the intermediate image and high-frequency information following a predetermined distribution, wherein the third resolution exceeds the first resolution and the input image and the output image have the same semantics.

Analyze

Prediction

Not rejected under §101

Rejection rate in neighbors: 9%

KNN search completed in 0.046s — using 11 nearest neighbors.

Suggested Changes

These suggestions are based on the words and bigrams present in the nearest claims in the embedding space. Positive weights indicate correlation with §101 rejection. Negative weights indicate correlation with no §101 rejection.

Consider Removing

Present in your claim and associated with rejection.

Token / Bigram	Weight	Percent Weight
image output	1.2335	37.0%
train	0.3674	11.0%
resolution input	0.3563	10.7%
output image	0.3230	9.7%
image train	0.3180	9.5%
distribution	0.3168	9.5%
high	0.2009	6.0%
method comprising	0.0883	2.6%
implement method	0.0640	1.9%
image high	0.0299	0.9%

Consider Adding

Not in your claim and associated with non-rejection.

Token / Bigram	Weight	Percent Weight
feature	-1.2874	8.9%
low	-1.0852	7.5%
perform	-0.9710	6.7%
set	-0.8260	5.7%
image process	-0.8150	5.6%
low resolution	-0.7944	5.5%
process image	-0.7034	4.8%
determine	-0.7032	4.8%
image generate	-0.7009	4.8%
generate image	-0.6893	4.7%

KNN hyperparameter selection

We evaluated K values from 3–15 and thresholds from 0.1–0.5. We chose K = 11 and threshold = 0.25 to balance overall accuracy with recall on the minority class (~12% of the data). The results we got:

- Accuracy: **84%**
- recall (no rejection): **88%**
- recall (rejection): **52%**

Future work

- Improve performance using ensemble models.
- Identify relevant Patent Trial and Appeal Board (PTAB) decisions and court rulings on claim validity using embeddings + KNN.